

04/27/2026

Exodromy & finishing the

proof that k_z is fully faithful

Last Time Compact generators for spectral presheaves,
Constructible Sheaves, the FLTZ Stratification

Today Combinatorial description of sheaves constructible
with respect to the FLTZ Stratification, finish the proof
that k_Z is fully faithful

Generalities on stratifications by affine hyperplane arrangements

> This is basically a recollection from last time

Def Let M be a lattice and $\mathcal{H}$ a finite set of hyperplanes in $M_{\mathbb{R}}$ that can be spanned by rational vectors. The periodic affine hyperplane arrangement associated to $\mathcal{H}$ is the collection of affine hyperplanes

$$\mathcal{H}^{\text{per}} = \{m + h \mid m \in M \text{ and } h \in \mathcal{H}\}$$

The depth of $\mathcal{H}$ is $\text{codim}_{M_{\mathbb{R}}}(\bigcap_{h \in \mathcal{H}} h)$

Ex Let (N, Σ) be a fan. The FLTZ periodic hyperplane arrangement $\mathcal{H}_{\Sigma}$ is the periodic hyperplane arrangement associated to the hyperplanes

$$\{\sigma^{\perp} \mid \sigma \in \Sigma(1)\}$$

Construction. Let M be a lattice and $\mathcal{H}$ a finite set of hyperplanes in $M_{\mathbb{R}}$. Define a stratification $M_{\mathbb{R}} \rightarrow P_{\mathcal{H}}$ by a poset of Krull dimension $n = \text{depth}(\mathcal{H})$ as follows.

Define a filtration $\text{Fil}^{\Sigma} M_{\mathbb{R}}$ on $M_{\mathbb{R}}$ as follows.

- > $\text{Fil}_n^{\Sigma}(M_{\mathbb{R}}) = M_{\mathbb{R}}$
- > $\text{Fil}_{n-1}^{\Sigma}(M_{\mathbb{R}}) = \bigcup_{h \in \mathcal{H}^{\text{per}}} h$
- > In general, $v \in M_{\mathbb{R}}$ is in $\text{Fil}_{n-i}^{\Sigma}(M_{\mathbb{R}})$ if and only if v belongs to at least i distinct hyperplanes in $\mathcal{H}^{\text{per}}$.

This gives a stratification over $\{0 < \dots < n\}$. We further refine the stratification by declaring that each connected component of $(M_{\mathbb{R}})_k$ is its own stratum. Then the set $P_{\mathcal{H}}$ of strata becomes a poset under the relation

$$S \leq T \text{ if and only if } S \subset \overline{T}.$$

Then when $\mathcal{H} = \{\sigma^\perp \mid \sigma \in \Sigma(1)\}$, we write $P_\Sigma = P_{\mathcal{H}}$ and refer to this stratification as the FLTZ stratification.

Recall. Let (N, Σ) be a fan. Then for each $\sigma \in \Sigma$ and $Z \in \Theta(\sigma)$, the sheaf $\omega_Z \in \text{Sh}(M_{\mathbb{R}})$ is constructible with respect to the FLTZ stratification.

Monodromy

- > In order to show that constructible sheaves for the FLTZ stratification is compactly generated (with an explicit set of compact generators), we want a generalization of the following to constructible sheaves.

Monodromy Thm. Let X be a locally weakly contractible space and $\mathcal{E}$ a presentable ∞ -category. There is a natural equivalence

$$\mathrm{LC}^{\mathrm{hyp}}(X, \mathcal{E}) \xrightarrow{\sim} \mathrm{Fun}(|X|, \mathcal{E})$$

locally constant hypersheaves

given on objects by

$$L \longmapsto [x \longmapsto L_x]$$

- > Let's try to understand a generalization.

Ex. Consider the stratification of $[0,1]$ by $\{0 < 1\}$ given by $\{0\}$ and $(0,1]$. What is $\text{Cons}_{\{0 < 1\}}([0,1], \mathcal{E})$?

> To describe locally constant sheaves on $\{0\}$ or $(1,0]$, we only need a stalk

> For $F \in \text{Cons}_{\{0 < 1\}}([0,1])$, we have a natural map

$$F_0 = \text{Colim}_{n \rightarrow \infty} F([0, 1/n]) \longrightarrow \text{Colim}_{n \rightarrow \infty} F((0, 1/n)) \quad \begin{array}{l} \text{all rest.} \\ \text{are equivs} \end{array}$$

s.t.

$$F([0,1]) \simeq F_1$$

> One can show that this data completely determines F . Namely, there is an equivalence

$$\text{Cons}_{\{0 < 1\}}([0,1], \mathcal{E}) \simeq \text{Fun}(0 \rightarrow 1, \mathcal{E}).$$

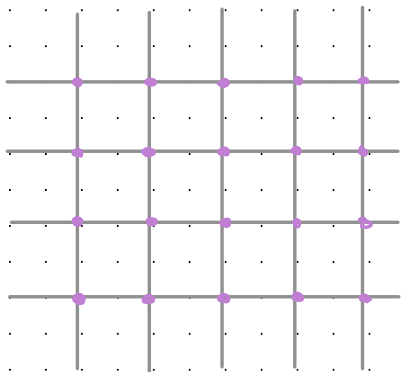
Ex. Let $N = \mathbb{Z}$, and consider the FLTZ Stratification of $M_{\mathbb{R}} \simeq \mathbb{R}$ determined by the Cone $\sigma = \mathbb{R}_{\geq 0}$



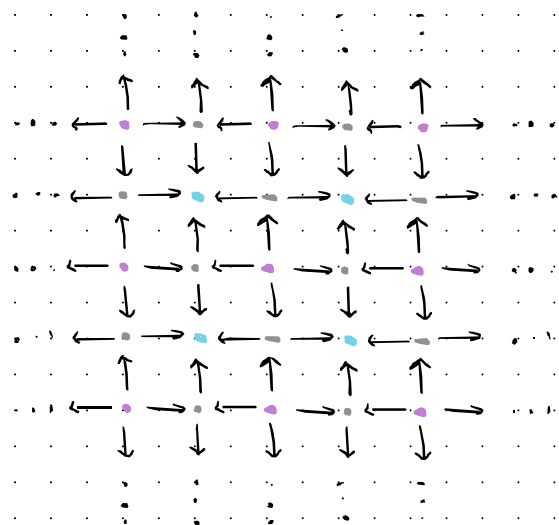
It is not difficult to bootstrap the above to see that Sheaves constructible for the FLTZ Stratification are functors out of the poset



Ex. Similarly, let $N = \mathbb{Z}^2$, then Σ be the cone generated by e_1 and e_2 , and consider the FLTZ Stratification of $M_{\mathbb{R}} \simeq \mathbb{R}^2$



It is not difficult to convince yourself that sheaves constructible with respect to this stratification are functors out of the poset



A simple case of exodromy

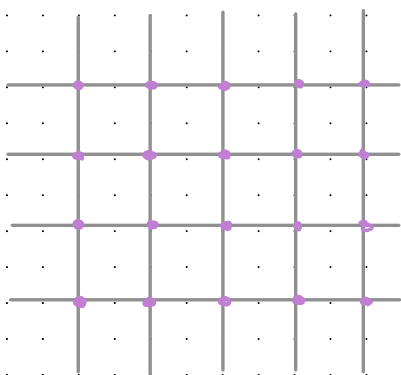
> Our next goal is to explain what happens in general.

Ntn Let $s: X \rightarrow P$ be a stratified space. The open star around p is

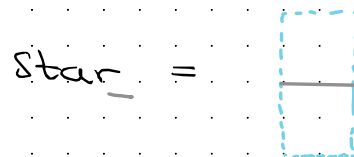
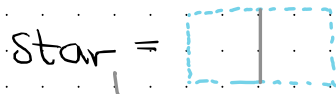
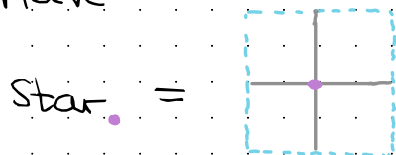
$$X_{\geq p} := s^{-1}(P_{\geq p})$$

This is also sometimes written star_p .

Ex. In the periodic hyperplane arrangement



We have



Def Let $f, f' : (X, P) \longrightarrow (Y, Q)$ be maps of stratified spaces. A stratified homotopy between f and f' is a stratified map

$$h : (X \times [0, 1], P) \longrightarrow (Y, Q)$$

where $X \times [0, 1]$ is stratified by $X \times [0, 1] \xrightarrow{pr_1} X \longrightarrow P$,

such that $h|_{X \times \{0\}} = f$ and $h|_{X \times \{1\}} = f'$

We say that f is a stratified homotopy equivalence if there is a stratified map $g : (Y, Q) \longrightarrow (X, P)$ such that gf and fg are stratified homotopy equivalent to the identity.

Baby exodromy theorem Let $s : X \longrightarrow P$ be a stratified top space. Assume that s is surjective and P satisfies the ascending chain condition. Also assume that there exists a collection B of open subsets of X such that

- (1) The sheaves represented by $U \in B$ generate $Sh(X, \mathcal{A}_n)$ under colimits.
- (2) For each $p \in P$, there exists an open $U \in B$ such that $U \subset X_{\geq p}$ and this inclusion is a stratified homotopy equiv.

Then for every compactly generated presentable ω -category $\mathcal{E}$, the functor

$$s^* \text{Fun}(P, \mathcal{E}) \longrightarrow \text{Sh}(X, \mathcal{E})$$

is fully faithful, preserves limits (and colimits), and has image $\text{Cons}_P(X, \mathcal{E})$.

If $\mathcal{E} = \text{Sp}$, then for each $p \in P$, the object $1_{x \geq p} = j_{x \geq p, !}(1)$ is compact and the smallest stable subcategory closed under colimits and containing these objects is all of $\text{Cons}_P(X, \text{Sp})$.

Remark In this case, the poset P is the exit-path ω -category of the stratified space (X, P) . The term 'exodromy' is a portmanteau of 'exit-path' and 'monodromy'. The above result is a special case of a much more general result.

- we won't prove this result, but we'll explain how to apply it in our situation of interest.

Prop Let M be a lattice and $\mathcal{H}$ a finite collection of hyperplanes in $M_{\mathbb{R}}$. Then the stratification $(M_{\mathbb{R}}, \mathcal{P}_{\mathcal{H}})$ determined by the periodic hyperplane arrangement $\mathcal{H}^{\text{per}}$ satisfies the conditions of the previous result.

Proof

For a point $x \in M_{\mathbb{R}}$ and $r > 0$, write $D(x, r) \subset M_{\mathbb{R}}$ for the open ball of radius r centered at x . Consider

$$\mathcal{B} = \left\{ D(x, r) \mid \begin{array}{l} x \in M_{\mathbb{R}} \text{ and } r > 0 \text{ is such that} \\ D(x, r) \text{ is contained in } \text{Star}_{S(x)}, \\ \text{and the inclusion is a stratified} \\ \text{homotopy equivalence} \end{array} \right\}$$

To verify that the hypotheses are satisfied, it suffices to show that $\mathcal{B}$ is a basis for the topology on X . In fact, we prove the stronger claim that for each $x \in X$, there is a radius $r_x > 0$ such that for $r \leq r_x$, the inclusions $D(x, r) \hookrightarrow D(x, r_x) \hookrightarrow \text{Star}_{S(x)}$ are stratified homotopy equivalences.

For this, observe that if we choose r_x small enough, then the only strata intersecting $D(x, r_x)$ are those containing x . Hence $(D(x, r_x), P_\Sigma)$ is homeomorphic to an $\mathbb{R}$ -vector space with a hyperplane arrangement. For this, it is clear that the inclusions $D(x, r) \hookrightarrow D(x, r_x)$ are stratified deformation retractions.

Also notice that this choice of r_x guarantees that $D(x, r_x) \subset \text{Star}_{s(x)}$. Since $\text{Star}_{s(x)}$ is convex, the straight-line homotopy provides the desired stratified homotopy equivalence. $\square$

Compact objects of constructible sheaves

Def. A poset P is locally finite if for each $p \in P$, the set $P_{\geq p}$ is finite.

Ex. If P is the poset of strata in a periodic hyperplane arrangement, then P is locally finite.

Lem. Let P be a finite poset and $\mathcal{C}_0$ a small ∞ -category. Then the inclusion $\text{Fun}(P, \mathcal{C}_0) \hookrightarrow \text{Fun}(P, \text{Ind}(\mathcal{C}_0))$ extends to an equivalence

$$\text{Ind}(\text{Fun}(P, \mathcal{C}_0)) \xrightarrow{\sim} \text{Fun}(P, \text{Ind}(\mathcal{C}_0))$$

In other words, if $\mathcal{C}$ is compactly generated, then

$$\text{Fun}(P, \mathcal{C})^\omega \simeq \text{Fun}(P, \mathcal{C}^\omega)$$

Lem. Let P be a locally finite poset. Then $F \in \text{Fun}(P, \mathcal{S}p)$ is compact if and only if F is nonzero on finitely many elements of P and each of its values is compact.

Proof.

As explained last time, $F \in \text{Fun}(P, \text{Sp})$ is compact if and only if F is in the smallest stable subcategory closed under retracts and containing the image of the spectral Yoneda embedding. Since P is locally finite, for each $p \in P$, the functor $\sum_+^\infty \text{Map}_P(p, -)$ is nonzero on only finitely many objects, and each value is a compact spectrum (specifically S).

Since the full subcategory of objects with compact values and nonzero on finitely many objects is stable and closed under retracts we deduce that F is in this subcategory.

Conversely, if F is in this subcategory, let

$$\text{supp}^*(F) = \{q \in P \mid \exists p \leq q \text{ such that } F(p) \neq 0\}$$

By assumption, $\text{supp}^*(F)$ is finite. Thus by the previous result,

$$F|_{\text{supp}^*(F)} \in \text{Fun}(\text{supp}^*(F), \text{Sp}^\omega) = \text{Fun}(\text{supp}^*(F), \text{Sp})^\omega$$

To conclude, observe that F is left Kan-extended from $\text{Supp}^*(F)$. Moreover, since the right adjoint to left Kan extension preserves colimits, left Kan extension preserves compact objects. $\square$

Completing the proof that k_Z is fully faithful

Lem. Let $N = \mathbb{Z}$ and let Σ be the fan spanned by $e_1, \dots, e_n$. Then

$$\text{Cons}_{P_\Sigma}(M_R) \subset \text{Sh}(M_R)$$

is closed under convolution and contains the unit for convolution.

Proof.

Recall that the unit for convolution is $O_*(S)$. Since $0 \in M_R$ is a closed stratum for the FLTZ stratification, it is immediate that $O_*(S)$ is constructible.

For closure under convolution, recall that $\text{Cons}_{P_\Sigma}(M_R)$ is the smallest subcategory of $\text{Sh}(M_R)$ closed under colimits and shifts and containing $\mathbb{1}_{\text{star}_p}$ for $p \in P_\Sigma$. Since the convolution product preserves colimits and shifts separately in each var, it suffices to show that for all $p, q \in P_\Sigma$, the convolution

$$\mathbb{1}_{\text{star}_p} * \mathbb{1}_{\text{star}_q}$$

is constructible with respect to the FLTZ stratification.

For this, observe that for each $p \in P_\Sigma$, the star star_p is the interior of a hyper-rectangle in $M_{\mathbb{R}}$ whose vertices are lattice points. Hence, by our earlier computations,

$$\mathbb{1}_{\text{star}_p} * \mathbb{1}_{\text{star}_q} \simeq \mathbb{1}_{\text{star}_p + \text{star}_q} [-\dim(M_{\mathbb{R}})]$$

Note that $\text{star}_p + \text{star}_q$ is also a hyper-rectangle with vertices in M . Hence

$$\mathbb{1}_{\text{star}_p + \text{star}_q} \in \text{Cons}_{P_\Sigma}(M_{\mathbb{R}}).$$

Since $\text{Cons}_{P_\Sigma}(M_{\mathbb{R}})$ is closed under shifts, $\mathbb{1}_{\text{star}_p} * \mathbb{1}_{\text{star}_q} \in \text{Cons}_{P_\Sigma}(M_{\mathbb{R}})$, as desired. $\square$

 For a general fan (N, Σ) , the subcategory $\text{Cons}_{P_\Sigma}(M_{\mathbb{R}})$ of $\text{Sh}(M_{\mathbb{R}})$ need not be closed under convolution.

Prop. Let (N, Σ) be a smooth fan. Then for each $\sigma \in \Sigma$, the functor

$$\Psi_{\sigma} : \text{Fun}(\Theta(\sigma)^{\circ}, Sp) \longrightarrow \text{Mod}_{\omega_{\sigma^{\vee}}} \text{Sh}(M_{\mathbb{R}})$$

is fully faithful. Hence so is Ψ_{Σ} and

$$K_{\Sigma} : \text{Qcoh}(X_{\Sigma}/T) \longrightarrow \text{Sh}(M_{\mathbb{R}})$$

Proof.

Since σ is smooth, there is a transformation in

$$SL(N) \simeq SL_n(\mathbb{Z})$$

that takes σ to $\text{Cone}(e_1, \dots, e_k)$ in the fan generated by $e_1, \dots, e_n \in \mathbb{Z}^n$. So without loss of generality, we can assume that Σ is this standard fan and $\sigma = \text{Cone}(e_1, \dots, e_k)$.

As explained last time, the functor $\Psi_{\sigma} : \text{Fun}(\Theta(\sigma)^{\circ}, Sp) \longrightarrow \text{Sh}(M_{\mathbb{R}})$ factors through $\text{Cons}_{P_{\Sigma}}(M_{\mathbb{R}})$. Since $\omega_{\sigma^{\vee}} = \Psi_{\sigma}(\sigma^{\vee})$ and $\text{Cons}_{P_{\Sigma}}(M_{\mathbb{R}})$ is closed under convolution product and contains the unit, we deduce that Ψ_{σ} factors through

$$\text{Mod}_{\omega_{\sigma^v}} \text{Cons}_{P_{\Sigma}}(M_{\mathbb{R}}) = \text{Cons}_{P_{\Sigma}}(M_{\mathbb{R}}) \cap \text{Mod}_{\omega_{\sigma^v}} \text{Sh}(M_{\mathbb{R}}).$$

> Observe that $\text{Mod}_{\omega_{\sigma^v}} \text{Cons}_{P_{\Sigma}}(M_{\mathbb{R}})$ is the smallest subcategory of $\text{Cons}_{P_{\Sigma}}(M_{\mathbb{R}})$ closed under colimits and shifts and containing

$$\omega_{\sigma^v} * 1_{\text{star}_p} \simeq \omega_{\sigma^v, 0} * 1_{\text{star}_p} \simeq 1_{\sigma^v, 0} + \text{star}_p$$

for $p \in P_{\Sigma}$. Moreover, these objects are compact in $\text{Mod}_{\omega_{\sigma^v}} \text{Cons}_{P_{\Sigma}}(M_{\mathbb{R}})$.

- To see this, note that

$$\omega_{\sigma^v} * (-) : \text{Cons}_p(M_{\mathbb{R}}) \rightarrow \text{Mod}_{\omega_{\sigma^v}} \text{Cons}_{P_{\Sigma}}(M_{\mathbb{R}})$$

is left adjoint to the inclusion, and both of these functors preserve colimits.

> Note that since $\sigma = \text{Cone}(e_1, \dots, e_k)$, we have

$$\sigma^v = \text{Cone}(e_1^*, \dots, e_k^*, \pm e_{k+1}^*, \dots, \pm e_n^*),$$

> Note that by the definition of the Minkowski sum, if R is the interior of a hyper-rectangle, then

$$\sigma^{v,0} + R = \sigma^{v,0} + m,$$

where m is the minimal vertex of $\overline{R}$.

- So $\sigma^{v,0} + \text{Star}_p = \sigma^{v,0} + m_p$, where m_p is the minimal vertex of $\overline{\text{Star}_p}$.

> Observe that as p varies, m_p runs over all elements of M .

- Fix $m \in M$ and let p be the lattice point (closed stratum) $m + (1, \dots, 1)$. Then Star_p is the interior of the hypercube of side length 2 centered at p . Hence m is the minimal vertex of $\overline{\text{Star}_p}$.

Upshot. For each $m \in M$, the object

$$\mathbb{1}_{\sigma^{v,0} + m} \in \text{Mod}_{\omega_{\sigma^v}} \text{Cons}_{\Sigma}^p(M, \mathbb{R})$$

is compact. Hence

$$\Psi_\sigma(m + \sigma^v) \cong \omega_{m + \sigma^v, 0} \cong 1_{m + \sigma^v, 0}[n]$$

is compact.

Conclusion. The functor

$$\Psi_\sigma: \text{Fun}(\mathcal{O}(\sigma)^{\text{op}}, \text{Sp}) \longrightarrow \text{Mod}_{\omega_\sigma} \text{Cons}_{\mathbb{P}_\Sigma}(M_{\mathbb{R}})$$

carries objects in the image of the spectral Yoneda embedding to compact objects. Moreover, for F and G in the image of Spectral Yoneda,

$$\underline{\text{Map}}(F, G) \longrightarrow \underline{\text{Map}}(\Psi_\sigma(F), \Psi_\sigma(G))$$

is an equivalence. From the results of last time, this proves that Ψ_σ is fully faithful. $\square$